12.602

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Question:
If the following system has a non-trivial solution:

$$px + qy + rz = 0,$$
$$qx + ry + pz = 0,$$
$$rx + py + qz = 0,$$

then which one of the following options is **TRUE**? (CS 2016)

(a) $p + q + r = 0$ or $p = q = r$
(b) $p + q + r = 0$ or $p = q = -r$
(c) $p + q + r = 0$ or $p = q = r = 0$
(d) $p + q + r = 0$ or $p = -q = r$

Solution:
Given equations are

$$\begin{pmatrix} p & q & r \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \tag{1}$$

$$\begin{pmatrix} q & r & p \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \tag{2}$$

$$\begin{pmatrix} r & p & q \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \tag{3}$$

$$\implies \begin{pmatrix} p & q & r \\ q & r & p \\ r & p & q \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{4}$$

let

$$\mathbf{A} = \begin{pmatrix} p & q & r \\ q & r & p \\ r & p & q \end{pmatrix} \text{ and } \mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \tag{5}$$

$$\implies \mathbf{AX} = \mathbf{0} \tag{6}$$

The system of equations $\mathbf{AX} = \mathbf{0}$ has non trivial solution if rank of $\mathbf{A} < 3$

$$\begin{pmatrix} p & q & r \\ q & r & p \\ r & p & q \end{pmatrix} \xleftrightarrow{R_2 \to pR_2 - qR_1} \begin{pmatrix} p & q & r \\ 0 & pr - q^2 & p^2 - qr \\ r & p & q \end{pmatrix} \tag{7}$$

$$\xleftrightarrow{R_3 \to pR_3 - rR_1} \begin{pmatrix} p & q & r \\ 0 & pr - q^2 & p^2 - qr \\ 0 & p^2 - rq & pq - r^2 \end{pmatrix} \tag{8}$$

$$\xleftrightarrow{R_3 \to (pr - q^2)R_3 - (p^2 - rq)R_2} \begin{pmatrix} p & q & r \\ 0 & pr - q^2 & p^2 - qr \\ 0 & 0 & D \end{pmatrix} \tag{9}$$

$$\implies D = (pr - q^2)(pq - r^2) - (p^2 - rq)(p^2 - qr) = 0 \tag{10}$$

$$\implies (pr - q^2)(pq - r^2) = p^2qr - pr^3 - pq^3 + q^2r^2 \tag{11}$$

$$\implies (p^2 - rq)(p^2 - qr) = p^4 - 2p^2qr + r^2q^2 \tag{12}$$

From (11) and (12)

$$D = \left(p^2qr - pr^3 - pq^3 + q^2r^2\right) - \left(p^4 - 2p^2qr + r^2q^2\right) \tag{13}$$

$$= -p^4 + 3p^2qr - pq^3 - pr^3 \tag{14}$$

$$= -p\left(p^3 + q^3 + r^3 - 3pqr\right) \tag{15}$$

$$= -p\,(p + q + r)\left(p^2 + q^2 + r^2 - pq - qr - pr\right) \tag{16}$$

$$\implies D = -\frac{1}{2}p\,(p + q + r)\left((p - q)^2 + (q - r)^2 + (p - r)^2\right) \tag{17}$$

$D = 0$ means

$$p + q + r = 0 \qquad \text{or} \qquad (p-q)^2 + (q-r)^2 + (p-r)^2 = 0 \tag{18}$$

$$\implies p + q + r = 0 \qquad \text{or} \qquad p = q = r \tag{19}$$

$\implies$ Given system of equations has non trivial solution if

$$p + q + r = 0 \qquad \text{or} \qquad p = q = r \tag{20}$$